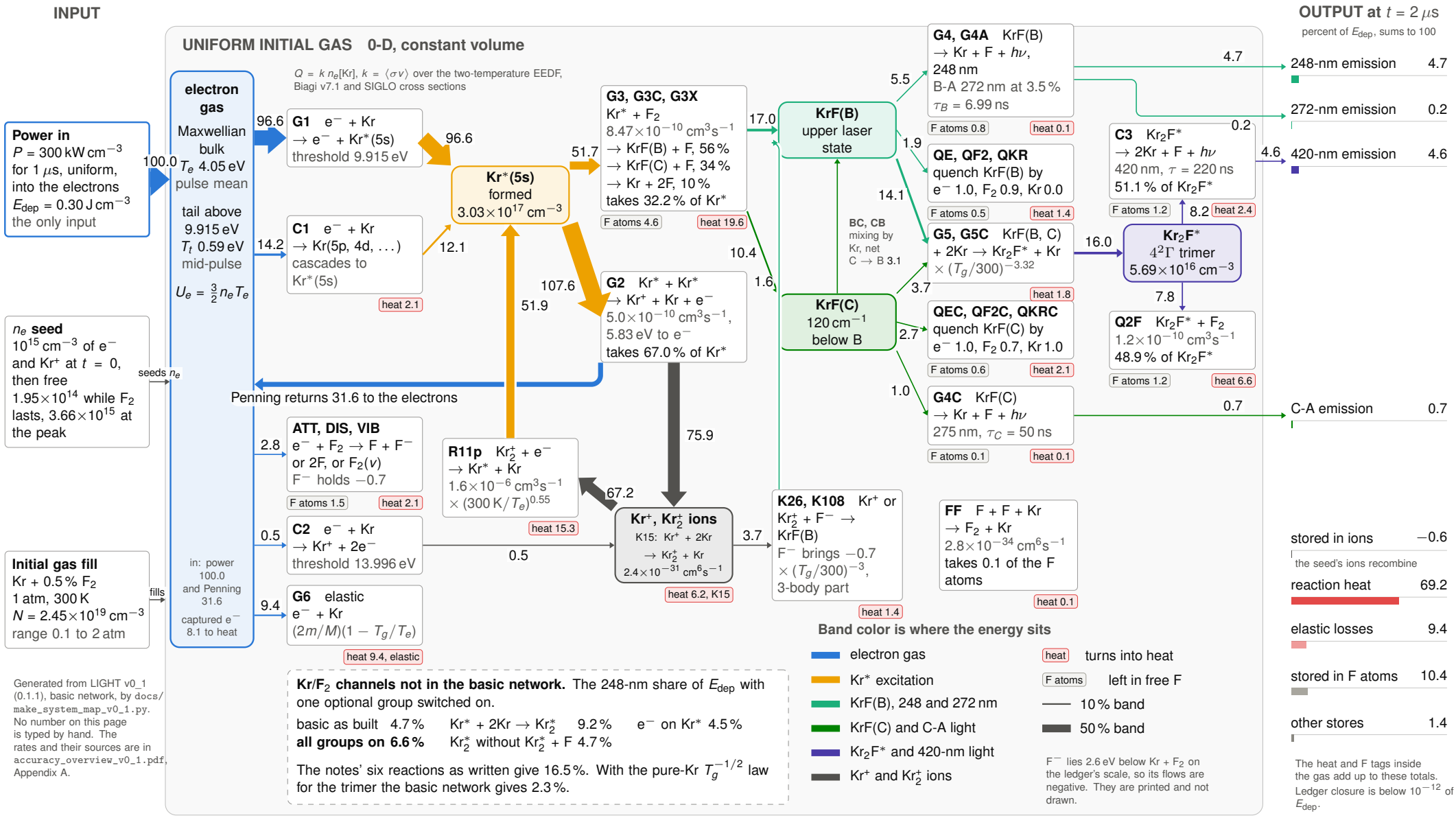
LIGHT system map, v0_1: energy flow through the Kr/F_2 kinetics

Reference point: 1 atm of Kr with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then free, run to 2 μs. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The notes' outputs as flows, which overlap: to Kr* 110.8 %, to Kr$^+$ 76.5 %, to KrF(B) and KrF(C) 24.8 %, heat 69.2 %, elastic losses 9.4 %, emission 10.2 %. They pass 100 % because recombination hands the ions' energy back to Kr*. Of the ions that recombine, 95.0 % meet an electron and return to Kr*, and 5.0 % meet F^- and form KrF(B). Of the KrF formed, 9.30×10^{16} cm^{-3}, 5.6 % came from ion-ion recombination. Of it 19.1 % radiates at 248 nm, 0.7 % at 272 nm and 3.3 % in the C-A band, 61.2 % forms the trimer, and electrons, F_2 and Kr quench 6.8 %, 5.6 % and 3.4 %. At the end of the run T_g = 762 K and p = 2.55 atm, and F_2 fell below 10 % of the fill at 613 ns.